

VARIANTS

A math slide has  
a seam. It just  
is not where the  
variant names are.

Every math slide after this one is at  
`size: portrait`, where the split fires.

Eight variants render **four** structures,  
so there are four preprocessors — and  
two scaffolds that keep the whole slide,  
because they have nothing to slice.

# What each structure does when the box gets tall

Equation and legend →

# What each structure does when the box gets tall

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- Equation and legend
  - The equation rides every page; the symbols page under it. A legend page without its equation is unreadable.

# What each structure does when the box gets tall (cont.)

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- Step table
  - The heading opens the run, then one proof step per page — each equation back on one line at full size.

# What each structure does when the box gets tall (cont.)

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- Card stack
  - Definition, Theorem and Proof were three cards crushed into one column. Now they are three pages.

# What each structure does when the box gets tall (cont.)

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- Columns
  - One labeled formulation per page, full measure, instead of two half-width columns stacked.

# Equation and legend: the symbols page under the equation.

The equation, then its  
symbols





# Equation and legend: the symbols page under the equation.

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$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$  — OLS coefficient vector

X →



# Equation and legend: the symbols page under the equation. (cont.)

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$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $X$  — design matrix,  $n \times p$

$y \rightarrow$

# Equation and legend: the symbols page under the equation. (cont.)

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$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $y$  — response vector, length  $n$

[Gram matrix →](#)

# Equation and legend: the symbols page under the equation. (cont.)

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$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $X^{\top} X$  — Gram matrix,  $p \times p$ , must be invertible

**The equation that  
was too wide for the  
slide, broken across  
lines.**

The equation, then its  
symbols



# The equation that was too wide for the slide, broken across lines.

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$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log (1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- $\ell$  — log-likelihood, concave in  $\beta$

the logistic link →

# The equation that was too wide for the slide, broken across lines. (cont.)

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$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log(1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- $\sigma$  — the logistic link

$y_i \rightarrow$

# The equation that was too wide for the slide, broken across lines. (cont.)

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$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log (1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- $y_i$  — observed label,  $\in \{0, 1\}$



# The equation that was too wide for the slide, broken across lines. (cont.)

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$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log (1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- $x_i$  — feature vector for observation  $i$



CHAIN RULE · FOUR STEPS

**Step table: one step  
per page, each on a  
single line.**

The derivation, one step  
per page



# Step table: one step per page, each on a single line.

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$$f(x + h) - f(x) = f'(x) h + O(h^2)$$

**subtract**  $f(x)$

continues →

# Step table: one step per page, each on a single line.

(cont.)

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$$\frac{f(x+h)-f(x)}{h} = f'(x) + O(h) \quad \left| \quad \text{divide by } h \neq 0 \right.$$

take the limit →

# Step table: one step per page, each on a single line.

(cont.)

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$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x) \quad \left| \quad \text{take the limit} \right.$$

**Card stack: one card per page, so a proof is read, not skimmed.**

The statement, then its  
proof



# Card stack: one card per page, so a proof is read, not skimmed.

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**DEFINITION.** A function  $f : [a, b] \rightarrow \mathbb{R}$  is *continuous* on  $[a, b]$  if  $\lim_{x \rightarrow c} f(x) = f(c)$  for every  $c \in [a, b]$ .

Theorem →

# Card stack: one card per page, so a proof is read, not skimmed. (cont.)

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**THEOREM.** Let  $f$  be continuous on  $[a, b]$  and let  $y$  lie strictly between  $f(a)$  and  $f(b)$ . Then there exists  $c \in (a, b)$  with  $f(c) = y$ .

Proof →



**Card stack: one card per page, so a proof is read, not skimmed.** (cont.)

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**PROOF.** Set  $S = \{x \in [a, b] : f(x) < y\}$ .  $S$  is non-empty and bounded; let  $c = \sup S$ . Continuity at  $c$  forces  $f(c) = y$ .



ESTIMATORS · ASYMPTOTICS

**Columns: one  
formulation per  
page, at the full  
measure.**

One formulation at a time →

# Columns: one formulation per page, at the full measure.

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## ORDINARY LEAST SQUARES

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$$\hat{\beta}_{\text{OLS}} = (X^{\top} X)^{-1} X^{\top} y$$

Unbiased under exogeneity; minimum variance among linear unbiased estimators.

Ridge →

# Columns: one formulation per page, at the full measure. (cont.)

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## RIDGE

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$$\hat{\beta}_{\lambda} = (X^{\top} X + \lambda I)^{-1} X^{\top} y$$

Biased, lower variance. The penalty  $\lambda$  buys stability when  $X^{\top} X$  is ill-conditioned.

**A fixed scaffold keeps the whole slide: there is nothing to slice.**

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$$\hat{\beta} = 0.42 \pm 0.03$$

$$\begin{array}{l} 95\% \text{ CI: } [0.36, 0.48] \\ p < 0.001 \quad \cdot \quad n = 1,204 \end{array}$$

The estimate, its uncertainty and the reading are one statement in three parts. Cut them apart and each page says less than the whole did, so this variant is left whole and rings if it ever runs long.